A balanced Pólya urn without a positive quadratic large-deviation rate

Disproof of Conjecture 00000002114

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Abstract

The classical two-colour Pólya urn with one initial ball of each colour and replacement matrix I_2 is balanced. Its count law is uniform. We prove that two separated closed tails of the colour proportion have zero exponential decay at speed n. Consequently no function $c(x-m)^2$ with $c > 0$, for any centre m, can satisfy the closed-set upper bound of a large-deviation principle. This refutes the asserted implication from balance to positive definiteness. The finite count recursion and the asymptotic contradiction are both proved in Lean 4.

Model and scope

At each step a ball is selected uniformly, returned, and one more ball of its colour is added. Initially there is one red and one blue ball. In the standard convention, the replacement matrix is

$$R = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

An urn is balanced when the two row sums are equal [2, Section 1]. Here both sums equal 1, and the total number of balls after n draws is $n + 2$.

Let K_n be the number of red draws and $X_n = (K_n+1)/(n+2)$ the red proportion. We use the conventional speed-n large-deviation question for X_n. The conjecture imposes no irreducibility or strictly positive off-diagonal hypothesis; this standard balanced urn is therefore an admissible input. The result refutes the positive-definiteness assertion under balance. It does not assert that every balanced urn fails to have a large-deviation principle, or that the zero quadratic form is excluded by the first clause alone.

Lemma 1. *For every $n \geq 0$ and $0 \leq k \leq n$, $\mathbb{P}(K_n = k) = 1/(n + 1)$.*

Proof. Write $p_n(k) = \mathbb{P}(K_n = k)$, with value zero outside $0 \leq k \leq n$. The initial law is $p_0(0) = 1$. For $1 \leq k \leq n$, the two incoming transitions give

$$p_{n+1}(k) = p_n(k)\frac{n-k+1}{n+2} + p_n(k-1)\frac{k}{n+2} = \frac{1}{n+2}.$$

At the two endpoints only one term survives: $p_{n+1}(0) = p_n(0)(n+1)/(n+2)$ and $p_{n+1}(n+1) = p_n(n)(n+1)/(n+2)$. Both equal $1/(n+2)$. Induction proves the formula. □

The large-deviation obstruction

Set $F_- = [0, 1/4]$ and $F_+ = [3/4, 1]$. For $n \geq 2$, the event $K_n = 0$ implies $X_n \in F_-$, while $K_n = n$ implies $X_n \in F_+$. Lemma 1 therefore gives

$$\frac{1}{n+1} \leq \mathbb{P}(X_n \in F_\pm) \leq 1.$$

Taking logarithms and dividing by $n > 0$ yields

$$-\frac{\log(n+1)}{n} \leq \frac{1}{n} \log \mathbb{P}(X_n \in F_\pm) \leq 0.$$

Since $\log(n+1)/n \to 0$, both middle expressions tend to zero.

Theorem 1. *For every $c > 0$ and $m \in \mathbb{R}$, the function $I(x) = c(x-m)^2$ cannot be a speed-n large-deviation rate function for X_n, even when it is assigned $+\infty$ outside $[0, 1]$.*

Proof. Such a rate would satisfy, for every closed set F,

$$\limsup_{n\to\infty} \frac{1}{n} \log \mathbb{P}(X_n \in F) \leq -\inf_{x\in F} I(x).$$

If $m \leq 1/2$, then $x - m \geq 1/4$ on F_+, so $\inf_{F_+} I \geq c/16 > 0$. The displayed upper bound says $0 \leq -c/16$, a contradiction. If $m > 1/2$, then $m - x \geq 1/4$ on F_-, and the same contradiction follows using F_-. Both sets are contained in $[0, 1]$, so extending the rate by infinity elsewhere has no effect on the argument. □

Thus balance does not imply a positive quadratic rate, even after allowing an arbitrary centre. In particular the usual candidate $c(x - 1/2)^2$ is impossible. A positive definite quadratic form for centred red/blue proportions restricts to such a positive quadratic on their one-dimensional constraint line, so changing to the two-coordinate description does not remove this obstruction.

What is formally checked

The Lean definition `mass` is the actual forward count recursion, with the zero endpoint separated and natural subtraction used only for the incoming blue term. Impossible states have mass zero. The theorem `mass_formula` solves it for every n, k, and `mass_total` proves normalization. Event probabilities are finite sums of these nonnegative masses at the actual proportion $(k+1)/(n+2)$; `prob_unit` proves that $[0, 1]$ has probability one.

`right_log_limit` and `left_log_limit` prove the two real logarithmic limits, using the polynomial lower bound and the logarithm's asymptotic growth. The definition `ClosedUpperBound` writes the actual limsup/infimum upper bound restricted to nonempty closed sets in $[0, 1]$ with eventually positive probability. This is a necessary part of the usual large-deviation upper bound. Restricting to these sets avoids the zero-probability logarithm convention; both sets used in the proof have strictly positive probability for every $n \geq 2$.

The theorem `no_positive_quadratic` proves that this necessary condition fails for every $c > 0$ and every real m, using the actual infimum of the quadratic on the selected closed interval. The final `counterexample` combines that conclusion with the explicit replacement matrix's balance. No existence theorem for a full LDP, or full classification of all urn rate functions, is claimed or needed.

Reproduction and review

The project pins Lean 4.19.0 and Mathlib revision `c44e0c8ee63ca166450922a373c7409c5d26b00b`. From the submitted `lean/` directory, run

```
lake exe cache get
lake build
lake env lean -DwarningAsError=true Main.lean
```

The complete submission is freshly built. The axiom audit contains only Lean's standard `propext`, `Classical.choice` and `Quot.sound`; there are no proof placeholders, custom axioms or external computation oracles.

The supplementary `verify.py` uses Python's standard library and exact fractions. It reconstructs the Markov transition law through $n = 200$, and independently enumerates all colour histories through $n = 10$ with evolving ball counts. It checks the uniform masses, normalization, support and the two tail bounds. These finite checks are not used to justify the asymptotic claim; the general formula and limits are proved in Lean. Run `python verify.py`, and compile this source with `tectonic main.tex`.

The exact bilingual conjecture is preserved in `SOURCE.md`. Build, provenance and review records are included in `verification/`. Review was an adversarial self-review by the author; no independent reviewer or subagent was used.

References

[1] The Last Math Competition, Conjecture 00000002114. Exact source and upstream provenance included with this submission.

[2] C. Banderier, P. Marchal and M. Wallner, *Periodic Pólya urns and an application to Young tableaux*, authors' preprint, February 7, 2018, Section 1. Primary source. This reference supplies the standard urn and balance conventions; the counterexample calculations above are self-contained.